

Agenda

1. Announcements

- 7.1 & 7.2 HW due today
- Discussion Quiz 2 tomorrow over 7.1 & 7.2
- Bring 7.4 worksheet back to discussion tomorrow
- My office hours are only 12-1pm today

2. Partial Fraction Decomposition (Section 7.4)

Irreducible Quadratic Factors

Let $a > 0$ be a constant. Then we have

$$\int \frac{dx}{x^2 + a^2} = \int \frac{dx}{a^2(\frac{x^2}{a^2} + 1)} = \frac{1}{a^2} \int \frac{dx}{(\frac{x}{a})^2 + 1} = \frac{1}{a} \int \frac{\frac{1}{a} dx}{(\frac{x}{a})^2 + 1}$$

$$\begin{cases} u = \frac{x}{a} \\ du = \frac{1}{a} dx \end{cases} = \frac{1}{a} \int \frac{du}{u^2 + 1}$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan(u) + C = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

Example: $\int \frac{(x+2)}{x(x^2+4)} dx$

$$\int \frac{x+2}{x(x^2+4)(x+2)} dx$$

$$\frac{x+2}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{(x^2+4)} = \frac{A(x^2+4) + (Bx+C)x}{x(x^2+4)}$$

Step 1 use strategic values x to find as many coefficients as possible

$$\begin{cases} x+2 = A(x^2+4) + (Bx+C)x \\ \underline{x=0:} \quad 2 = 4A + \cancel{(Bx+C)(0)} \\ \quad \quad \quad \boxed{A = \frac{1}{2}} \end{cases}$$

Step 2 put in known coefficients, set common terms

$$x+2 = \frac{1}{2}(x^2+4) + (Bx+C)x$$

$$\underline{0}x^2 + \underline{x} + \underline{2} = \underline{\frac{1}{2}}x^2 + \underline{2} + \underline{B}x^2 + \underline{C}x$$

$$\begin{aligned} 0 &= \frac{1}{2} + B \rightarrow \boxed{B = -\frac{1}{2}} \\ 1 &= C \rightarrow \boxed{C = 1} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \int \frac{1}{x} - \frac{1}{2} \int \frac{x+1}{x^2+4} \\ &= \frac{1}{2} \int \frac{1}{x} dx - \frac{1}{2} \int \frac{1}{x^2+4} dx + \frac{1}{2} \int \frac{x}{x^2+4} dx \\ &= \frac{1}{2} \ln|x| - \frac{1}{4} \ln(x^2+4) + \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C \end{aligned}$$

$u = x^2 + 4$
 $du = 2x dx$

Activity: $\int \frac{2}{x(x^2+1)^2} dx$

$$\frac{2}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$\frac{2}{x(x^2+1)^2} = \frac{A(x^2+1)^2}{x(x^2+1)^2} + \frac{(Bx+C)x(x^2+1)}{x(x^2+1)^2} + \frac{(Dx+E)x}{x(x^2+1)^2}$$

$$2 = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x$$

$$x=0$$

$$(A=2)$$

$$0x^4 + 0x^3 + 0x^2 + 0x = 2x^4 + 4x^2 + 2 + Bx^4 + Bx^2 + Cx^3 + Cx + Dx^2 + Ex$$

$$0 = 2 + B$$

$$0 = C$$

$$0 = 4 + B + D$$

$$0 = C + E$$

$$\begin{cases} B = -2 \\ C = 0 \\ D = -2 \\ E = 0 \end{cases}$$

$$2 \int \frac{1}{x} - \int \frac{2x}{x^2+1} - \int \frac{2x}{(x^2+1)^2}$$

$u = x^2+1$
 $du = 2x dx$

$$2 \ln|x| - \ln(x^2+1) - \frac{1}{x^2+1} + C$$

We have a general form for partial fraction linear decomposition with linear and quadratic factors.

Example: Write out (but do not solve the partial fraction decomposition

$$\textcircled{1} \frac{3x^3 - 2x^2 + 5x + 1}{x(x+1)^2(x^2+1)(x^2+4)}$$

$$\textcircled{2} \frac{x}{(x-1)^3(x^2+9)^2(x^2+16)}$$

Activity $\int \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} dx$